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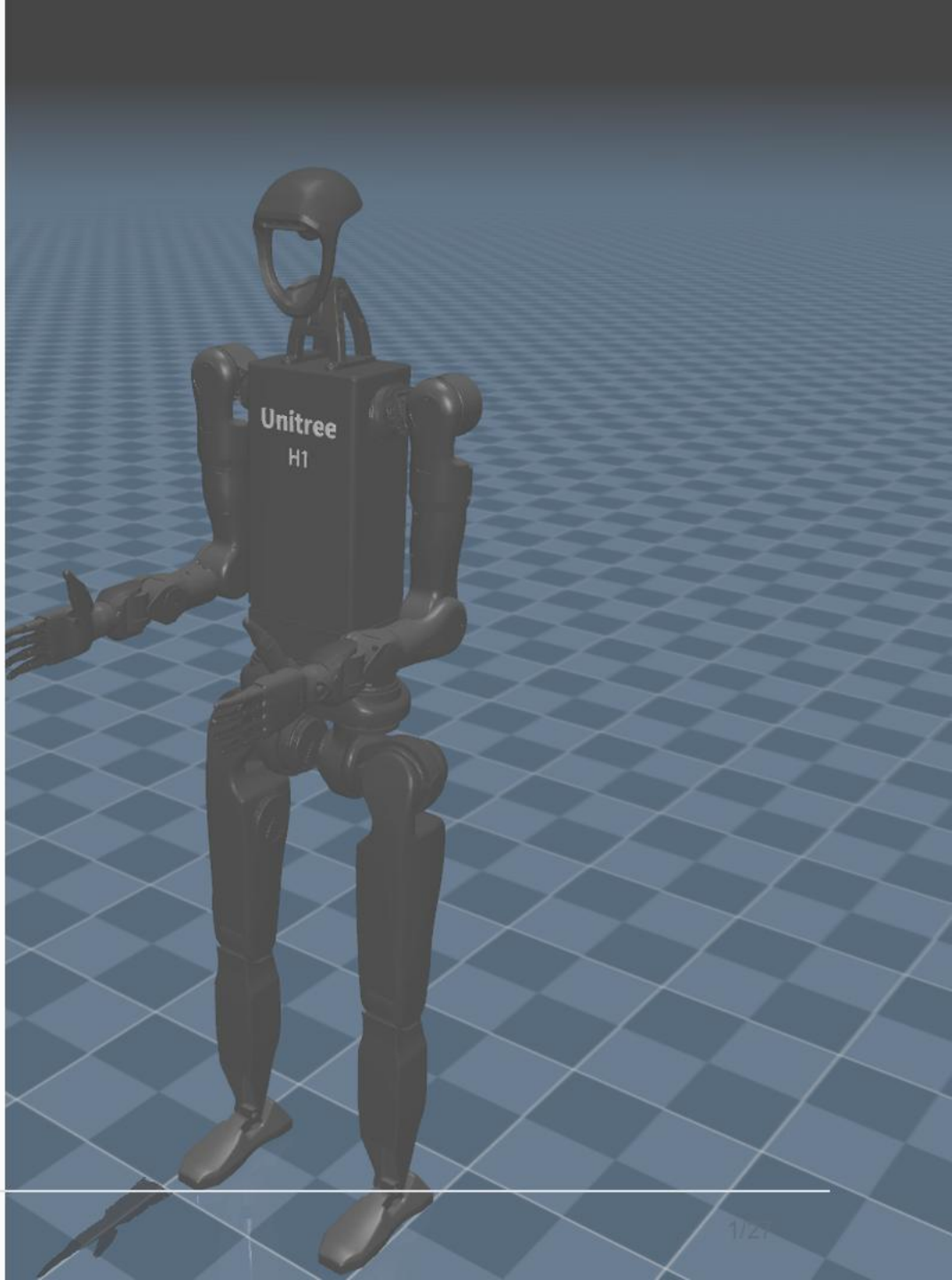
Unitree H1-2 Robot Telemetry, XR Control, and Digital Twin Platform

Initial visual executive presentation draft

Prepared for leadership review

Visual source: repository robot, model, teleoperation, and Vodafone assets.

Telemetry • XR • Digital Twin • Recording • Replay

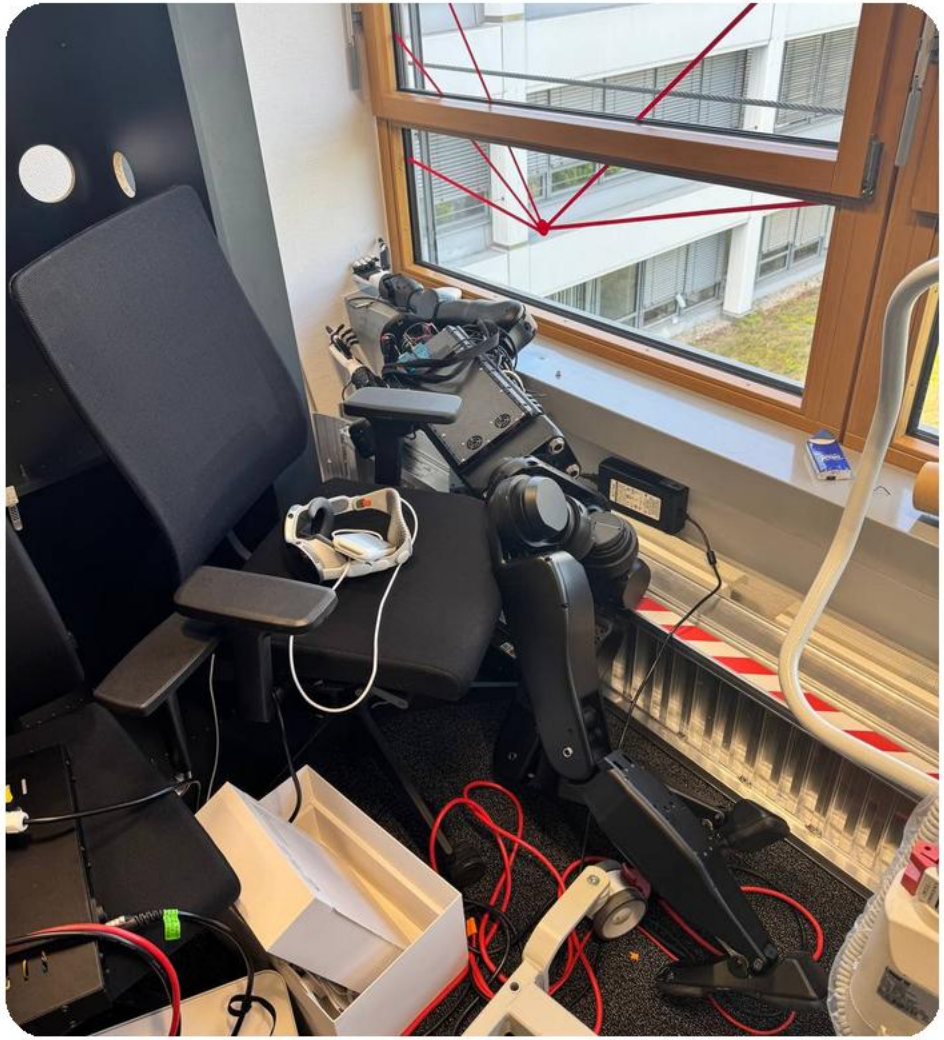


1. Executive Summary

This repository is a working operator platform for the Unitree H1-2 humanoid robot.

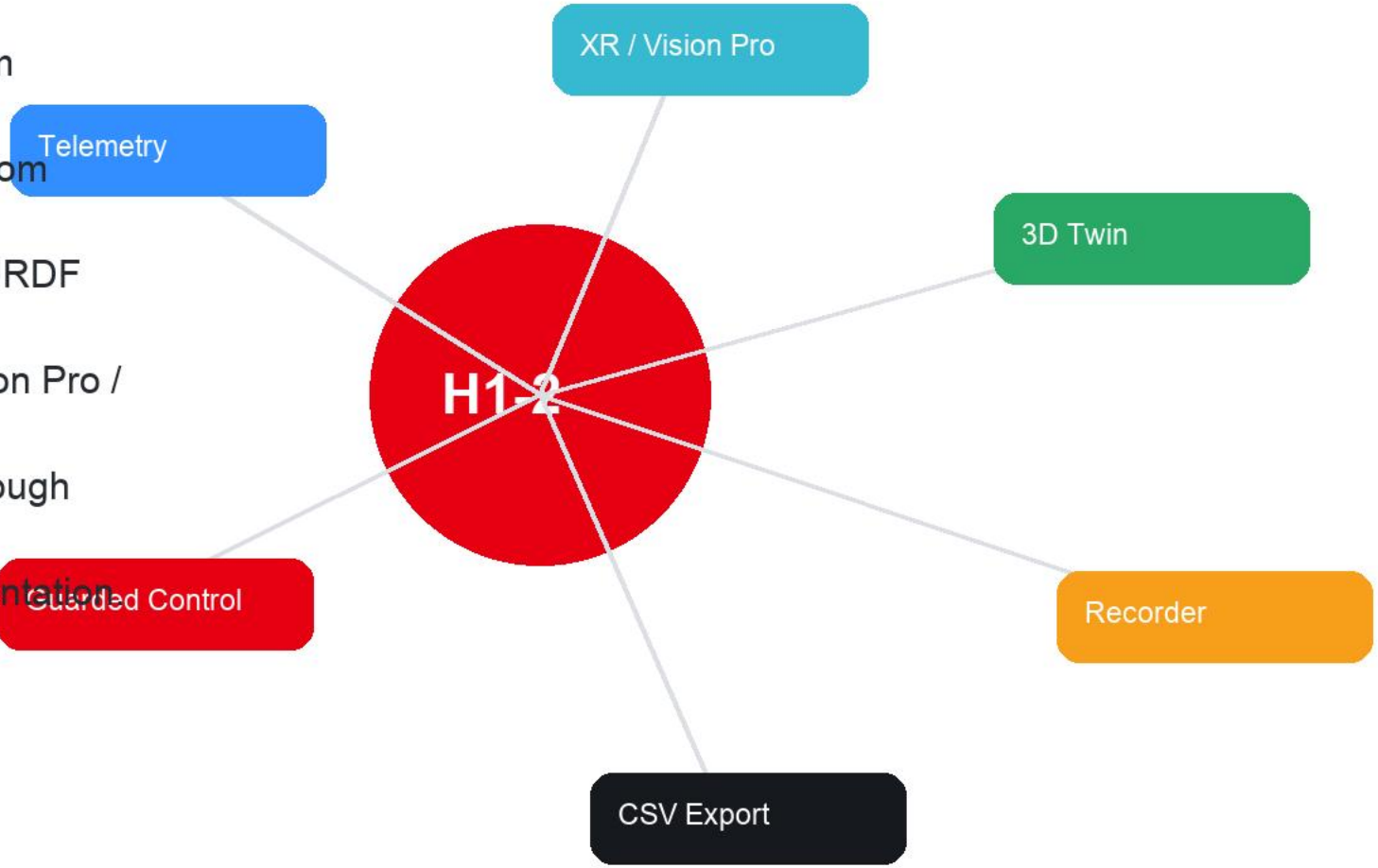
It combines a local web dashboard, live robot telemetry, guarded command endpoints, Vision Pro / XR teleoperation services, camera streaming, full-body telemetry recording, replay visualization, CSV export, deployment automation, and offline validation tooling.

The current system is already useful for supervised robot operation, operator demos, debugging, data capture, and early imitation-learning dataset preparation.



2. What The Platform Delivers

- Live browser dashboard for robot state and supervision.
- Unitree DDS telemetry ingestion from rt/lowstate.
- RH56BFX / Inspire hand telemetry from rt/inspire/state.
- 3D H1-2 digital twin rendered from URDF and STL assets.
- TeleImager camera access and Vision Pro / XR teleoperation entry points.
- Guarded H1 locomotion controls through Unitree LocoClient.
- Guarded right-wrist control experimentation.

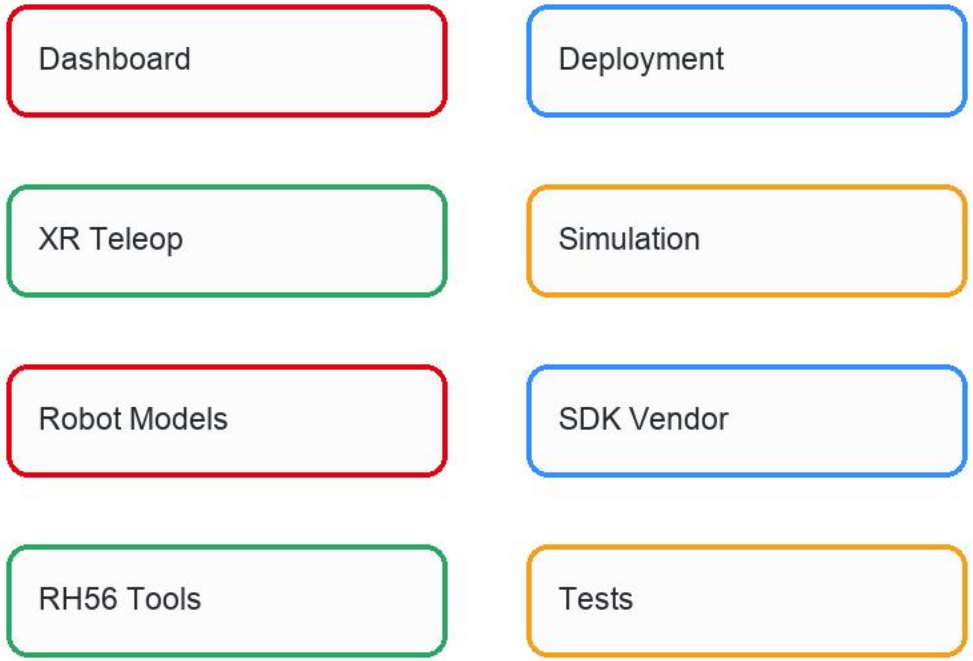


3. Repository Scope

The repository is more than a frontend project.

It consolidates:

- Dashboard server and static frontend.
- H1-2 robot model assets.
- Vision Pro / XR teleoperation integration.
- Robot deployment service files and patch scripts.
- Semantic teleoperation simulation and execution workspaces.
- Unitree SDK2 Python vendor snapshot.



4. High-Level Architecture

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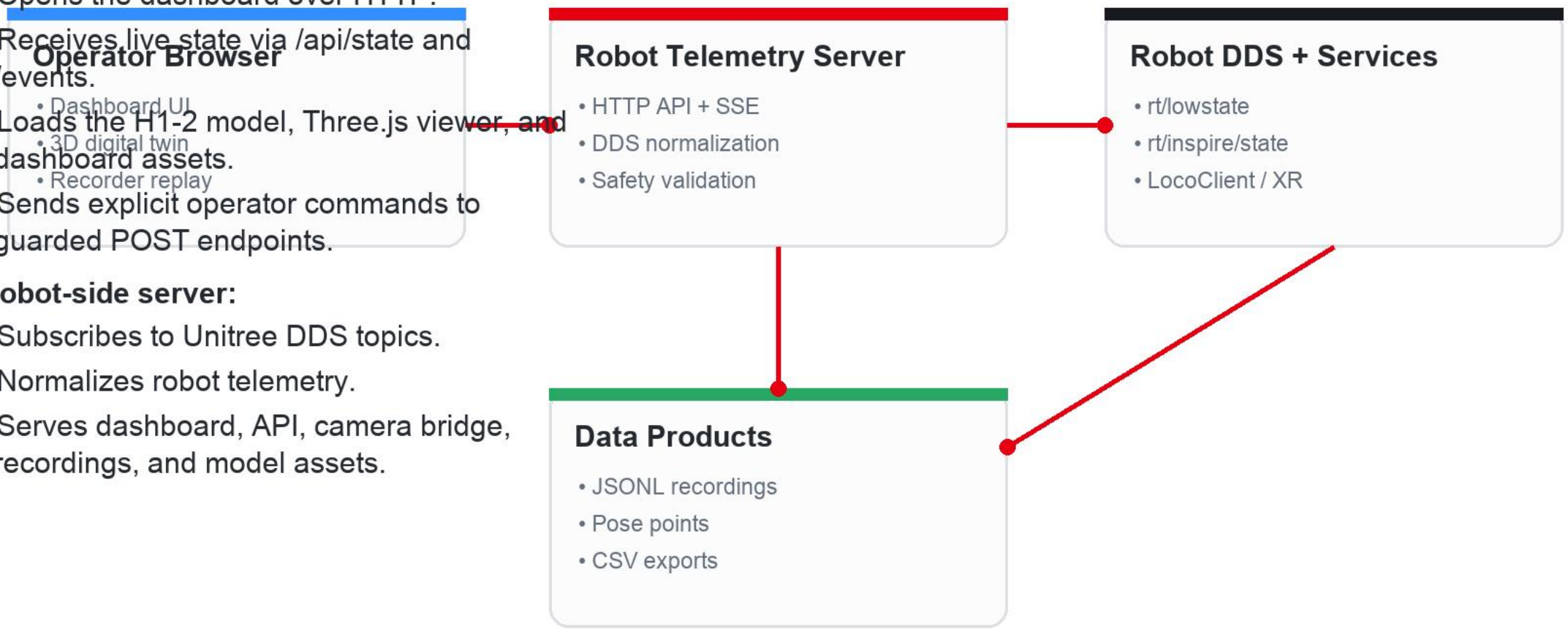
Operator browser:

- Opens the dashboard over HTTP.
- Receives live state via /api/state and /events.
- Loads the H1-2 model, Three.js viewer, and dashboard assets.
- Sends explicit operator commands to guarded POST endpoints.

Robot-side server:

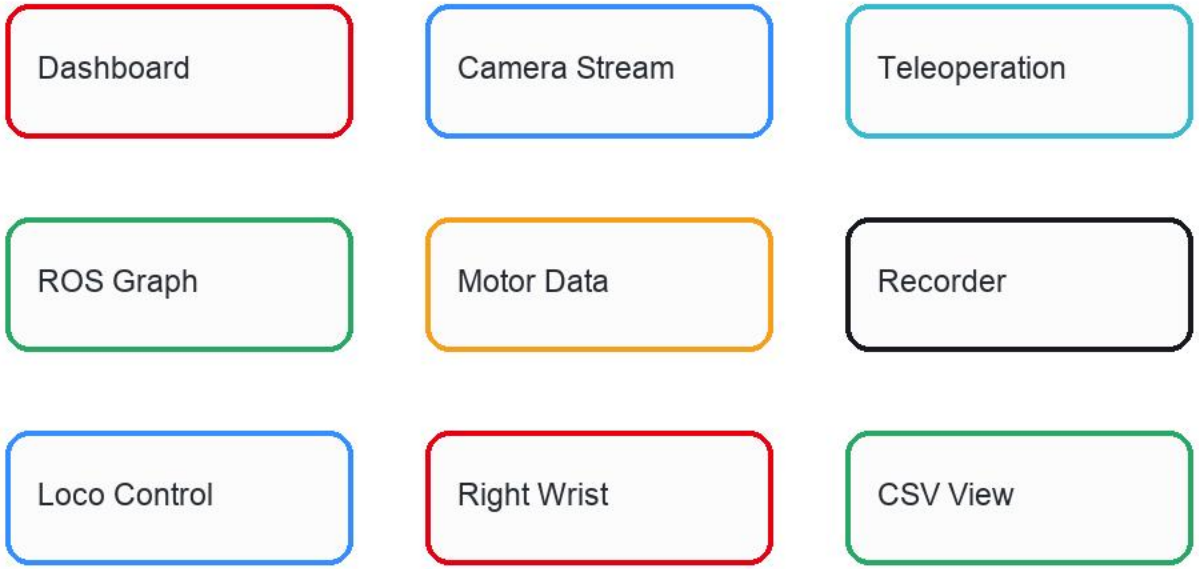
- Subscribes to Unitree DDS topics.
- Normalizes robot telemetry.
- Serves dashboard, API, camera bridge, recordings, and model assets.



5. Dashboard Pages

The current dashboard includes these operator pages:

- Dashboard
- Camera Stream
- Teleoperation
- ROS Graph
- Motor Data
- Recorder



6. Live Telemetry Features

The platform reads and displays:

- Connection state.
- Snapshot age.
- Sample rate.
- Total sample count.
- Motor count.
- Latest telemetry error.
- Robot metadata: version, mode_pr, mode_machine, tick, crc, and wireless remote state.
- IMU data: quaternion, gyroscope, accelerometer, roll / pitch / yaw, and temperature.
- Battery / BMS values when exposed by firmware.
- Foot force and estimated foot force arrays.
- RH56BFX / Inspire hand connection, sample count, joint count, and finger positions.

7. Full-Body Motor Coverage

The system has a named H1-2 motor contract for 27 body joints:

- Left leg: hip yaw, hip pitch, hip roll, knee, ankle pitch, ankle roll.
- Right leg: hip yaw, hip pitch, hip roll, knee, ankle pitch, ankle roll.
- Waist: waist yaw.
- Left arm: shoulder pitch, shoulder roll, shoulder yaw, elbow, wrist roll, wrist pitch, wrist yaw.
- Right arm: shoulder pitch, shoulder roll, shoulder yaw, elbow, wrist roll, wrist pitch, wrist yaw.

Reserved motor slots are handled explicitly as ReservedMotorSlot<N>.

8. Hand Telemetry Coverage

The RH56BFX / Inspire hand contract tracks 12 named hand joints:

- RightPinky
- RightRing
- RightMiddle
- RightIndex
- RightThumbBend
- RightThumbRotation
- LeftPinky
- LeftRing
- LeftMiddle
- LeftIndex
- LeftThumbBend

9. 3D Digital Twin Viewer

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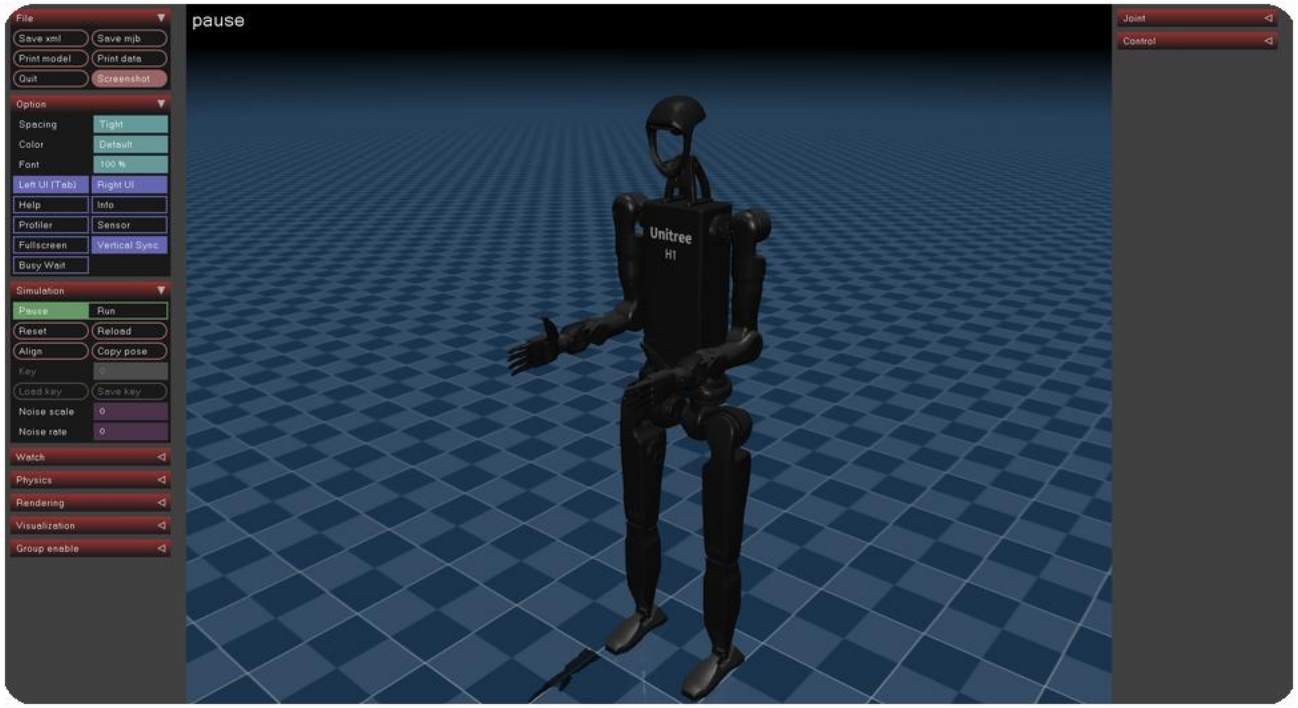
The live digital twin uses:

- static/models/h1_2_description/h1_2.urdf
- Unitree H1-2 STL mesh assets.
- Vendored Three.js modules.

Viewer capabilities:

- H1-2 mesh rendering.
- Body motor telemetry mapped to URDF joints.
- RH56BFX hand telemetry mapped to finger joints.
- Orbit and zoom controls.



URDF STL meshes Live joints

10. Camera And XR Experience

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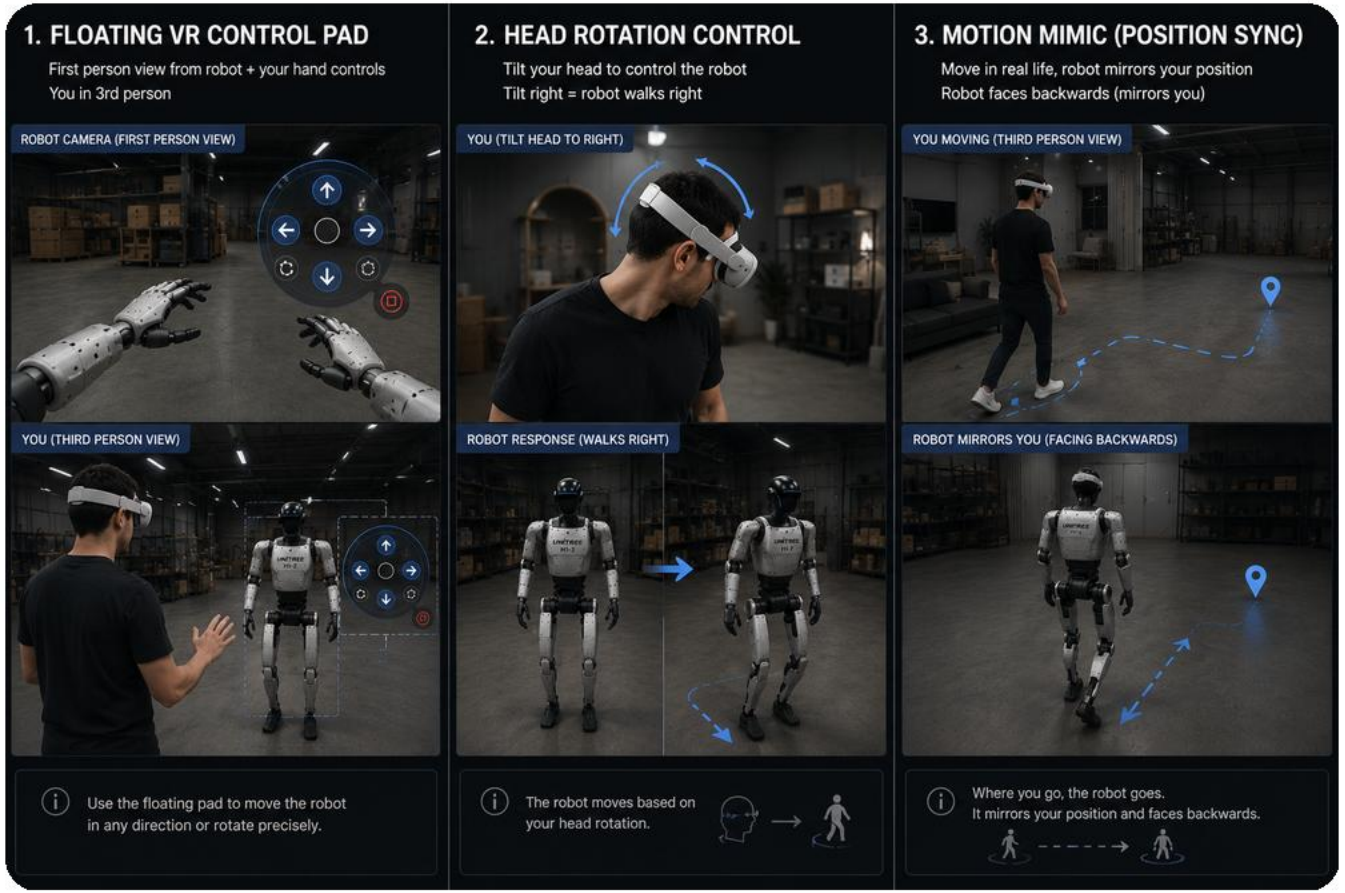
Camera features:

- Embedded TeleImager WebRTC preview.
- Direct camera page link.
- Certificate trust guidance for browser access.
- Camera backend state through /api/camera.
- Optional MJPEG bridge at /camera.mjpg.

XR features:

- Vision Pro / Vuer page link.
- Robot-hosted XR service on port 8012.



11. H1 Locomotion Controls

The dashboard exposes guarded Unitree H1 LocoClient controls:

- Ready / Balance Stand.
- Stand Up.
- Start.
- Stop Move.
- Damp.
- Zero Torque.
- High Stand / Low Stand.
- Set stand height.
- Set swing height.
- Velocity command.
- Move command.

12. Right-Wrist Control Surface

The right-wrist page focuses on RightWristYaw, motor index 26.

Capabilities:

- Live current q and dq readout.
- Absolute target command.
- Relative step command.
- Back-and-forth oscillation mode.
- Adjustable kp, kd, duration, period, and rate.
- Optional automatic gain selection.
- arm_sdk and lowcmd control path experimentation.
- Explicit arming and risk acknowledgement before command execution.
- Stop Wrist action.

This area is experimental and intentionally guarded.

13. Safety Model

The project includes code that can move a physical robot, so safety is treated as a product feature.

Current safeguards:

- Command validation on the server.
- Numeric limit checks for locomotion and wrist payloads.
- Fail-closed behavior when robot clients are unavailable.
- Explicit armed=true and i_understand_risk=true checks for wrist commands.
- Separate high-level loco path from high-risk low-level body control.
- Robot replay endpoint intentionally refuses physical playback until a safety controller exists.
- Production gate separates offline checks from live robot checks.
- README documents trusted-network requirements and unauthenticated HTTP risk.

14. Telemetry Recorder

The Recorder page supports two capture modes:

- Sequence recording.
- Pose point capture.

Sequence recording writes JSONL records under recordings/:

- recording_start
- telemetry_sample
- command_event

15. Replay And Comparison Digital Twin

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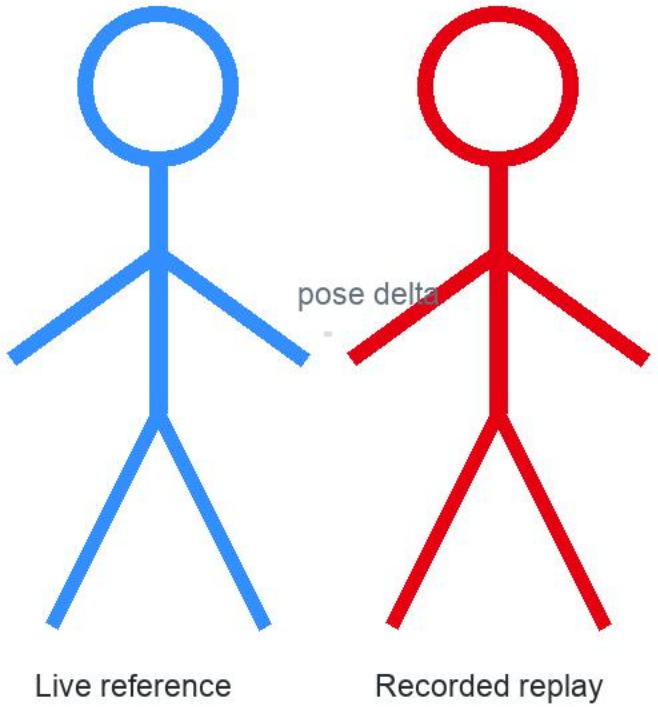
The Recorder page includes a replay viewer that is independent from the live robot command path.

Replay features:

- Recording file selector.
- Load, Play, Pause, scrub slider, and speed selector.
- JSONL sequence replay.
- Pose point replay with interpolation from live pose to target pose.
- Two-robot comparison:
- Translucent blue reference model from live robot state.



16. CSV And Data Analysis

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The dashboard includes a CSV View for the current telemetry snapshot.

The repository also includes tools/recording_to_csv.py, which converts recordings into analysis-ready CSV tables:

- samples.csv
- body_motors.csv
- hand_joints.csv
- imu.csv
- forces.csv

samples.csv

sample | timestamp | mode

body_motors.csv

motor | q | dq | tau

hand_joints.csv

hand | q | dq

imu.csv

quat | gyro | rpy

forces.csv

force | force_est

events.csv

time | command

17. ROS Graph And Control Discovery

The dashboard can expose ROS graph information through /api/ros-graph.

The repository documents current robot control paths, including:

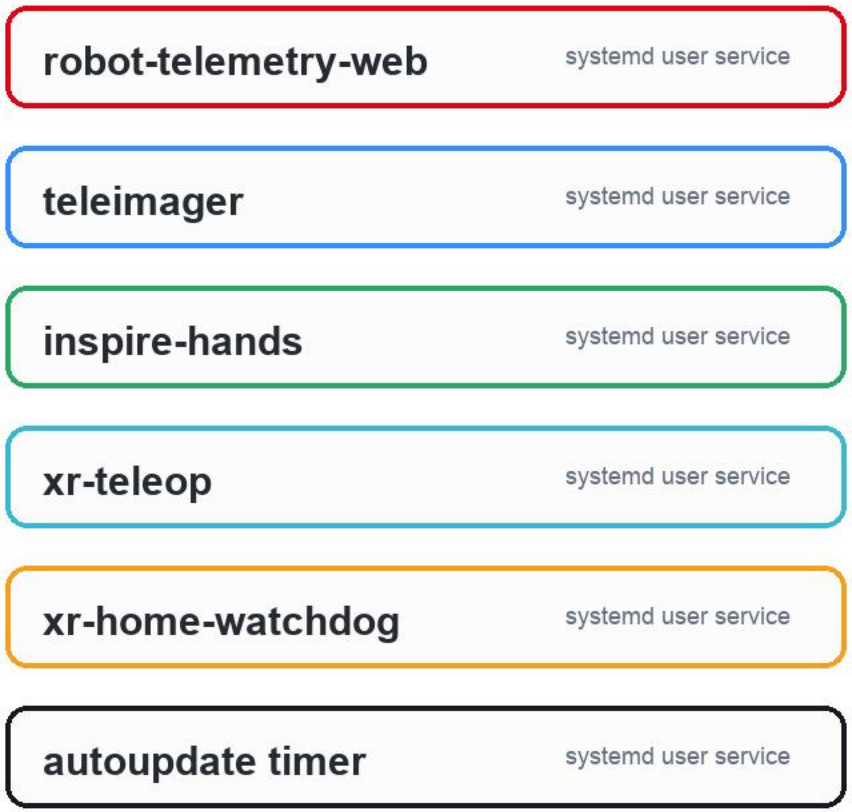
- /lowstate / rt/lowstate.
- /lowcmd / rt/lowcmd.
- /arm_sdk.
- /loco_sdk.
- /api/loco/request and /api/loco/response.
- /api/motion_switcher/request and response.
- Battery, mainboard, odometry, wireless controller, Inspire hand, video, and estimator topics.

This documentation is important because it prevents unsafe guessing when moving from observation to control.

18. Vision Pro / XR Deployment

The repository contains robot-side XR deployment automation:

- XR teleoperation checkout update.
- Vuer / XR launcher patching.
- Camera configuration patching.
- Image server patching.
- Dexterous retargeting patching.
- Inspire hand direct-curl patching.



19. Service Operations

Robot-side services:

- Dashboard server on port 8088.
- TeleImager camera server on port 60001.
- XR / Vuer service on port 8012.
- Inspire hand bridge.
- XR home watchdog.
- Auto-update timer from origin/main.

Operational helpers:

- run_servers.py starts the local dashboard.
- kill_servers.py stops local or robot-side dashboard processes.
- deployment/install_robot_services.sh installs the robot service set.
- deployment/robot_autoupdate.sh updates the robot checkout safely.

Logs are written under /home/unitree/logs.

20. Simulation And External Workspaces

The repository includes simulation and execution assets from the semantic teleoperation workspace:

- Unitree MuJoCo assets and scripts.
- Unitree ROS 2 support.
- Gazebo packages.
- MoveIt configuration.
- H1 visualization packages.
- H1 motion demo scripts.
- Real-robot guarded handshake script.

Vision Pro control also includes references to Unitree XR teleoperation and Isaac Lab simulation workflows.

This creates a path from offline replay and synthetic data into simulation validation before physical robot playback.

21. Developer And Test Tooling

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Quality and operations tooling includes:

- Unit / contract tests in tests/test_contracts.py.
- make test.
- make production-gate.
- Python syntax checks for owned scripts.
- Shell syntax checks for deployment scripts.
- JavaScript syntax checks for dashboard files.
- Contract checks for body joint ordering, hand joint ordering, wrist safety flags, command limits, loco actions, and recorder record ordering.

The production gate is intentionally offline by default, so it can run without robot access and without publishing robot commands.

22. Current Limitations

Important limitations are explicit:

- The dashboard does not implement authentication.
- It must run only on trusted robot networks.
- Local development without Unitree SDK / CycloneDDS can test UI but not live telemetry.
- Raw trajectory playback to the physical robot is intentionally locked.
- Physical replay needs interpolation, joint limits, velocity limits, torque limits, controller ownership checks, emergency stop supervision, and simulation validation.
- Some arm and motion-switcher ownership details still require live robot investigation.
- No explicit project license is currently included.

23. Business Value

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The platform gives the team:

- A visible operator experience for demos.
- A live debugging console for robot telemetry.
- A safer path to supervised command testing.
- A reusable data capture pipeline.
- Replay and comparison tools for movement analysis.
- CSV export for analytics and ML preparation.
- XR teleoperation deployment automation.

24. Recommended Next Milestones

Near-term:

- Add authentication or network access control.
- Add clearer operator safety state and E-stop integration.
- Add richer recording metadata and session labels.
- Add CSV download directly from the dashboard.
- Add more tests around telemetry normalization and recorder conversion.

Robot playback readiness:

- Build a dedicated replay safety controller.
- Validate in simulation first.
- Add interpolation and limits.
- Add controller ownership checks.
- Add emergency stop supervision.
- Add supervised small-range physical tests.

25. Roadmap Opportunities

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Potential feature extensions:

- Recording library with tags, notes, operator name, and robot configuration.
- Motion comparison metrics between live and replay pose.
- Dataset export for imitation learning.
- Automatic anomaly detection for joint temperature, torque, voltage, and sample-rate drops.
- Web-based CSV download and charting.
- Pose-to-pose planner with speed and smoothness controls.
- Multi-camera view support.
- Cloud sync for selected non-sensitive recordings.
- Role-based dashboard access.
- Live health checklist before enabling command pages.

26. Closing

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The project has moved from a telemetry viewer into a broader robot operator platform.

It now covers live state monitoring, XR operation, camera access, guarded control, full-body recording, digital twin replay, CSV data export, deployment automation, simulation assets, and offline validation.

The next major step is not adding raw power. It is adding production-grade safety, access control, and validated motion playback.